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Special Units and Euler Systems

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Abstract

The interaction of the units of a number field and its ideal class group has long been a topic of study in algebraic number theory. In the 1980s, Rubin discovered the notion of "special units" and it was conjectured that they encode information about the exponent of the class group. Expanding on these ideas, Kolyvagin introduced the machinery of Euler systems, and they were subsequently used by Rubin to provide a new proof of the Main Conjecture of Iwasawa theory. We use the machinery of Kolyvagin and Rubin to study the properties of special units and provide evidence for Rubin's conjecture.

Preliminaries

Motivation: Fermat's Last Theorem. For fixed integer $n > 2$, the equation $a^n + b^n = c^n$ has no integer solutions with $a, b, c \neq 0$.

Proof Attempt (Lamé)[2]: Assume such a solution exists; without loss of generality $n = p$ is an odd prime. Factor the equation in a suitable setting. i.e. factor in $\mathbb{Z}[\zeta_p]$:

$$a^p + b^p = (a+b)(a+b\zeta_p)\cdots(a+b\zeta_p^{p-2}) = c^n.$$

Use unique factorization in the appropriate ring to derive a contradiction.

Why the "Proof Attempt" is just an attempt: $\mathbb{Z}[\zeta_n]$ is in general not a unique factorization domain; the primes p for which $\mathbb{Z}[\zeta_p]$ has unique factorization are $\{3, 5, 7, 11, 13, 17, 19\}$. An example of failure of unique factorization is

$$(1+\sqrt{-5})(1-\sqrt{-5}) = 2 \cdot 3 = 6$$

in $\mathbb{Z}[\sqrt{-5}]$.

Fixing the Proof (Kummer): Replace unique factorization with something similar; Kummer called these "ideal numbers" but today we use "ideals" in the ring $\mathbb{Z}[\zeta_n]$ and use the fact that these ideals satisfy a unique factorization property.

Definition: Ideals. An *ideal* of a ring R is a subset $I \subset R$ such that for any $x, y \in I$ and $r \in R$, we have $x + y, -x, rx \in I$.

Using ideals in Kummer's approach: We view the expression

$$(a+b)(a+b\zeta_p)\cdots(a+b\zeta_p^{p-2}) = (c^n)$$

as an equality of ideals rather than elements. Since ideals factor uniquely in $\mathbb{Z}[\zeta_p]$, we can similarly derive a contradiction when p is a "regular prime".

Definition: Ideal Class Group. Form an equivalence relation on the set of ideals of $\mathbb{Z}[\zeta_n]$ by $I \sim J$ if $\alpha I = \beta J$ for some $\alpha, \beta \in \mathbb{Z}[\zeta_n]$. The principal ideals form their own equivalence class and serve as the identity for this group under the operation $[I][J] = [IJ]$. We denote this group by Cl_n.

Theorem: Trivial Class Groups. Cl_n is the trivial group if and only if $\mathbb{Z}[\zeta_n]$ is a unique factorization domain. For example, $\mathrm{Cl}_{\mathbb{Q}} = 1$ but $\mathrm{Cl}_{\mathbb{Q}(\zeta_{23})} \neq 1$.

Theorem: Finiteness of the Class Group. For any n, the ideal class group Cl_n is finite. We denote its size by h (or h_n).

Regular Primes: A prime p is called *regular* if $p \nmid h_p$. Examples of regular primes include $23, 29, 31$. Kummer showed through his use of ideals (or "ideal numbers" at the time) that Fermat's Last Theorem holds for exponent p when p is a regular prime. It is conjectured that there are infinitely many regular primes and that their density (proportion to all primes) is $e^{-1/2}$. There are infinitely many *irregular* primes such as $37, 59, 67$, so even with Kummer's method, Fermat's Last Theorem remains unresolved in its full generality.[a]

Rings of Integers: The integers $\mathbb{Z}$ have a special relationship with the rational numbers $\mathbb{Q}$: every element of $\mathbb{Q}$ is a quotient of two integers, and we can moreover talk about unique factorization in $\mathbb{Z}$. This allows us to in a way speak of factorization in $\mathbb{Q}$ as well e.g.

$$\frac{33}{50} = 2^{-1} \cdot 3 \cdot 5^{-2} \cdot 11.$$

In a similar vein, when we have any number field K –a subfield of $\mathbb{C}$ which is finite-dimensional as a $\mathbb{Q}$-vector space –we can form the *ring of integers* $\mathcal{O}_K$ of K. $\mathcal{O}_K$ has a similar relationship to K as $\mathbb{Z}$ has to $\mathbb{Q}$: every element of K is a quotient of elements in $\mathcal{O}_K$, and in certain cases we can speak of unique factorization in $\mathcal{O}_K$. When $K = \mathbb{Q}(\zeta_n)$, $\mathcal{O}_K = \mathbb{Z}[\zeta_n]$. As with $\mathbb{Z}[\zeta_n]$, we define the ideal class group of $\mathcal{O}_K$ which we denote Cl_K. This group is still finite, and we denote its size by h_K. This is called the *class number* of K.

[a] After 358 years of mathematical efforts, including the development of the fields of algebraic number theory and algebraic geometry, Fermat's Last Theorem was finally resolved in 1994 with Andrew Wiles' proof of the Modularity Theorem for semistable elliptic curves.

Extending Tools: How to study Cl_n?

The Relative Class Number[5]: For $K = \mathbb{Q}(\zeta_n)$, the field $K^+ = K \cap \mathbb{R}$ is the unique *real* subfield of K over which K is a 2-dimensional vector space. Denote by h_n and h_n^+ the class numbers of K and K^+, respectively. Then $h_n^+ \mid h_n$; we denote the quotient by $h_n^- := h_n/h_n^+$. This is called the *relative class number*, and it is well understood, so studying Cl_n amounts to studying Cl_{K^+}.

$$\begin{array}{ccc} K & = \mathbb{Q}(\zeta_n) & \mathrm{Cl}_K \\ 2 \Big| & & \Big| h_K^- \\ K^+ & = K \cap \mathbb{R} & \mathrm{Cl}_{K^+} \\ \frac{\varphi(n)}{2} \Big| & & \Big| h_K^+ \\ \mathbb{Q} & & \mathrm{Cl}_{\mathbb{Q}} = 1 \end{array}$$

Introducing Units: An alternative approach to defining/understanding the class group is through the following *exact sequence*:

$$1 \longrightarrow \mathcal{O}_K^\times \longrightarrow K^\times \longrightarrow J_K \longrightarrow \mathrm{Cl}_K \longrightarrow 1.$$

This encodes the following: we consider the subgroup of "principal ideals" $P_K \leq J_K$, which we can canonically identify with $K^\times/E_K$; then $\mathrm{Cl}_K = J_K/P_K$. This shows that we can try to gain understanding of Cl_K through understanding E_K. Our first results on E_K also show the difficulty in understanding Cl_{K^+}:

Dirichlet's Unit Theorem: For K a number field, we write r_1 for the number of embeddings of K into $\mathbb{R}$ and r_2 for the number of *pairs* of embeddings of K into $\mathbb{C}$ (so $r_1 + 2r_2 = [K : \mathbb{Q}]$, the dimension of K as a $\mathbb{Q}$-vector space). Then the *units* of $\mathcal{O}_K$ (multiplicatively invertible elements) have the following group structure:

$$E_K = \mathcal{O}_K^\times \cong W_K \times \mathbb{Z}^{r_1+r_2-1}$$

where W_K is a finite abelian group. The consequence for real fields is that the unit group is larger relative to the index $[K : \mathbb{Q}]$ than for imaginary fields.

Relating Units to the Ideal Class Group: Cyclotomic Units. We denote by C_{K^+} the subgroup of E_{K^+} generated by ± 1 and norms of elements of the form $1 - \zeta_n^k$. Then we have the following due to Sinnott[4]:

$$[E_{K^+} : C_{K^+}] = h_K^+ M_K$$

where M_K is a constant (dependent on K^+) that is explicitly computable. In some cases, for example when K^+ is a quadratic field, we have $[E_{K^+} : C_{K^+}] = h_K^+$. However, the groups Cl_{K^+} and E_{K^+}/C_{K^+} are in general not isomorphic –E_{K^+}/C_{K^+} will be cyclic but Cl_{K^+} need not be.

Generalizing: Special Units. For any abelian extension of number fields K/F, we can define a filtration of subgroups of E_K known as the *special units*; when $F = \mathbb{Q}$, all special units contain the cyclotomic units, and when F is an imaginary quadratic field all special units contain the elliptic units[?]. This leads us to the following conjecture[3] for $F = \mathbb{Q}$:

$$\mathrm{Cl}_K \cong \mathrm{gr}(S_K) \quad \text{as } \mathbb{Z}'[\mathrm{Gal}(K/\mathbb{Q})] \text{ modules}$$

where $\mathrm{gr}(S_K)$ is a graded module constructed from the special units and $\mathbb{Z}'$ is a localization of $\mathbb{Z}$ at certain "poorly behaved" primes.

Main Result

The main result of this thesis is the following reformulation and strengthening of Lemma 2.6 in [3]:

Definitions: Let K be a real abelian field, write $M = p^{2\nu_p(h_K)}$, and let I_p be the set of rational primes l, along with 1, such that $l \equiv 1 \bmod M$ and l splits completely in K. Let χ denote a p-adic valued Dirichlet character of $\mathrm{Gal}(K/\mathbb{Q})$ and for any $\mathbb{Z}[\mathrm{Gal}(K/\mathbb{Q})]$-module M, let $M(\chi)$ denote the χ-eigenspace of $M \otimes_{\mathbb{Z}} \mathbb{Z}_p$; there exists $\epsilon \in E_K$ such that ϵC_K generates $(E_K/C_K)(\chi)$. Let I_ϵ denote the subset of I_p such that any $l \in \mathrm{I}_\epsilon$ splits completely in the splitting field of $x^p - \epsilon$ and the class of some prime $\mathfrak{l}$ of K above l is contained in $\mathrm{Cl}_K(\chi)$.

Lemma: Let $l \in \mathrm{I}_p$ and write $l - 1 = p^{\nu_l} m_l$ where $(m_l, p) = 1$. For any prime $\mathfrak{l}$ of K over l, ϵ^{m_l} generates the p-Sylow subgroup of $(\mathcal{O}_K/\mathfrak{l})^\times$ if and only if l does not split completely over the splitting field of $x^p - \epsilon$.

Proposition: Set $W = \left\langle \epsilon^{p^s} \right\rangle \leq K^\times/(K^\times)^M$ for some $s \in \mathbb{N}$. Let l be a rational prime which splits completely in $K(\zeta_M)$ and is completely inert in the extension $K(\zeta_M, W^{1/M})$. For any prime $\mathfrak{l}$ of K over l, ϵ^{m_l} generates the p-Sylow subgroup of $(\mathcal{O}_K/\mathfrak{l})^\times$. Moreover, $\mathrm{I}_p \setminus \mathrm{I}_\epsilon$ is infinite.

Where to proceed from here?

- Seo's conjecture ($\mathrm{Cl}_K \cong \mathrm{gr}(S_K)$)
- $[E_K : S_K] = \exp(\mathrm{Cl}_K)$, where exp denotes the exponent of the group –this would follow from Seo's conjecture
- The current results on these conjectures are formulated through *Euler Systems* –a powerful tool discovered by Victor Kolyvagin[1] and used by Karl Rubin to prove the *Main Conjecture of Iwasawa Theory*[?]. Kolyvagin and Rubin also used Euler systems to make major progress on the *Birch and Swinnerton-Dyer Conjecture*, one of the Millennium Prize Problems.

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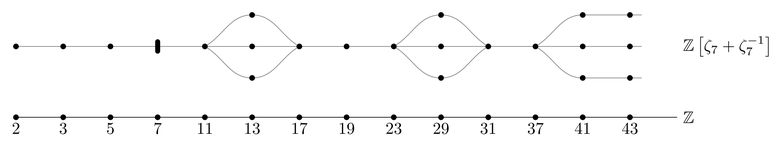

Figure 1. Visualization of the primes in $\mathcal{O}_{K^+}$ for $K = \mathbb{Q}(\zeta_7)$

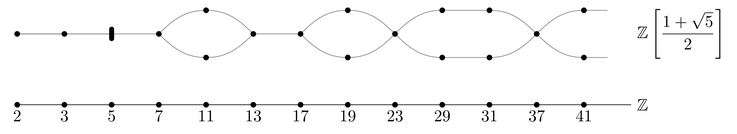

Figure 2. Visualization of the primes in $\mathcal{O}_{K^+}$ for $K = \mathbb{Q}(\zeta_5)$

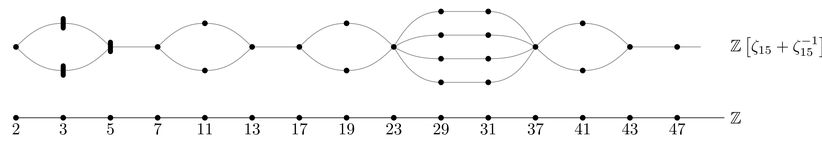

Figure 3. Visualization of the primes in $\mathcal{O}_{K^+}$ for $K = \mathbb{Q}(\zeta_{15})$